

Space-Filling Designs for Computer Experiments

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Design Objective

We want points $x_1, \dots, x_n \in [0, 1]^d$ such that:

- **Good coverage of the space**
- Avoid clustering
- Avoid large empty regions

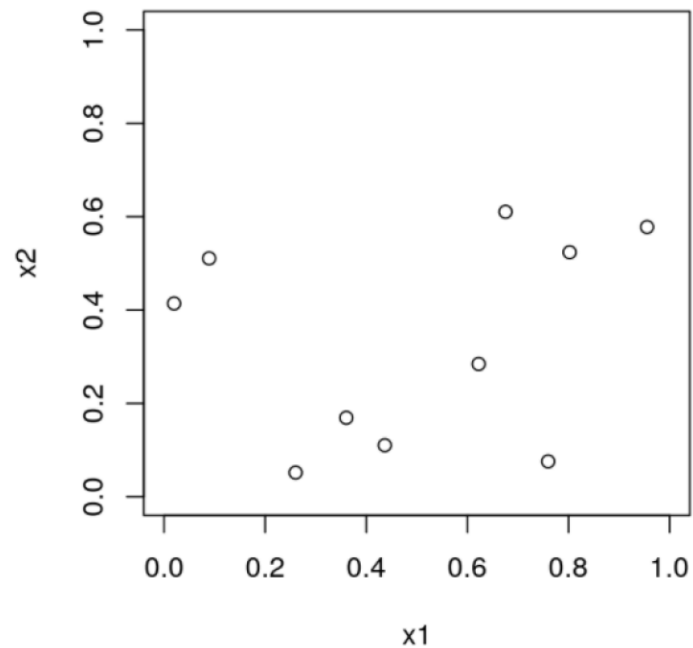
Two complementary ideas:

1. **Uniform coverage:** points spread evenly
2. **Separation:** points far from each other

These lead to different criteria

Why not simple random uniform?

```
p <- 2  
n <- 10  
X <- matrix(runif(n*p), ncol=p)
```



- The trouble: randomness is clumpy
We are almost guaranteed to get design locations right next to one another, and thus gaps in other parts of the input space.
- It may help a little if we do several replicates.
- Reasonable “fill” in the 2D input space is quite rare indeed.

Key Criteria

- Minimax distance
- Maximin distance
- Discrepancy

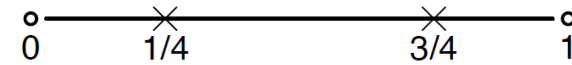
Minimax Criterion

Minimize worst-case distance to nearest point:

$$\min_{x \in [0,1]^d} \max_i \min \|x - x_i\|$$

- $p = 1, n = 2$:

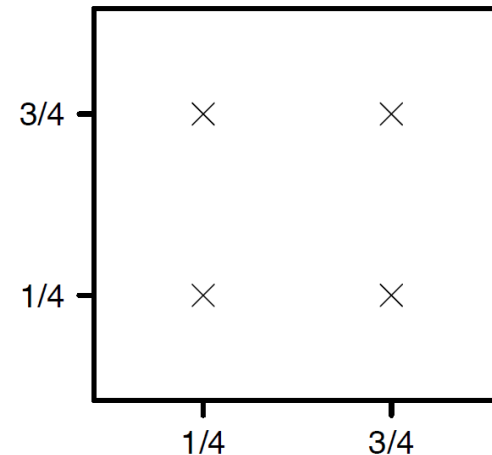
No region is too far from a design point



- $p = 2, n = 4$:

Interpretation

controls the largest “hole”



Maximin Criterion

Choose design to maximize:

$$\max \min_{i \neq j} \|x_i - x_j\|$$

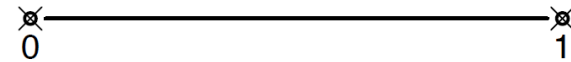
Points are as far apart as possible

Interpretation:

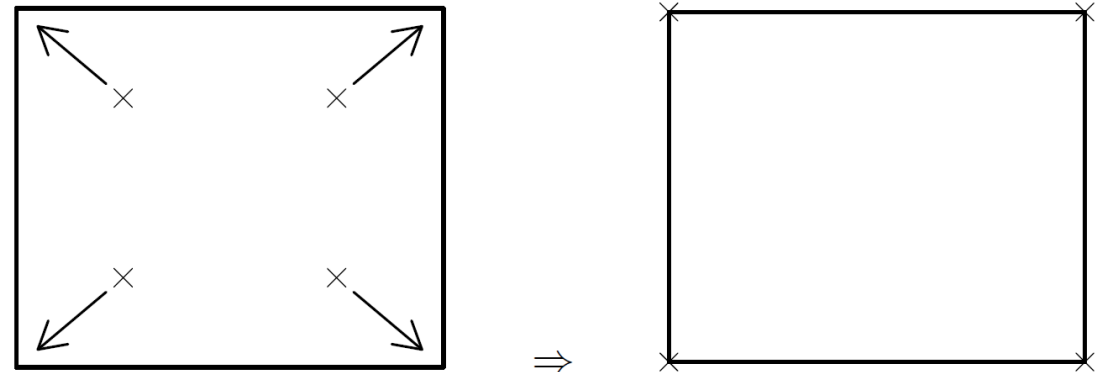
avoids clustering

improves global coverage

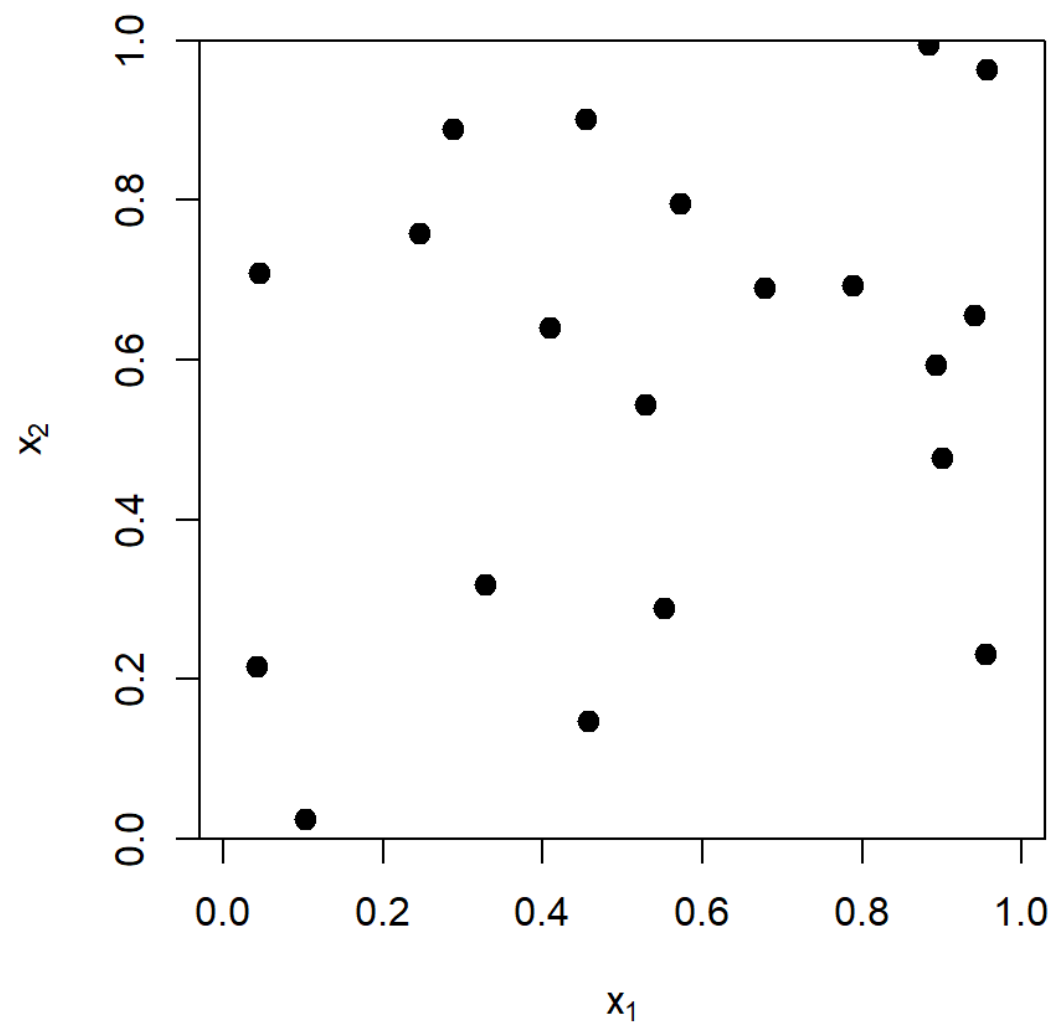
- $p = 1, n = 2$:



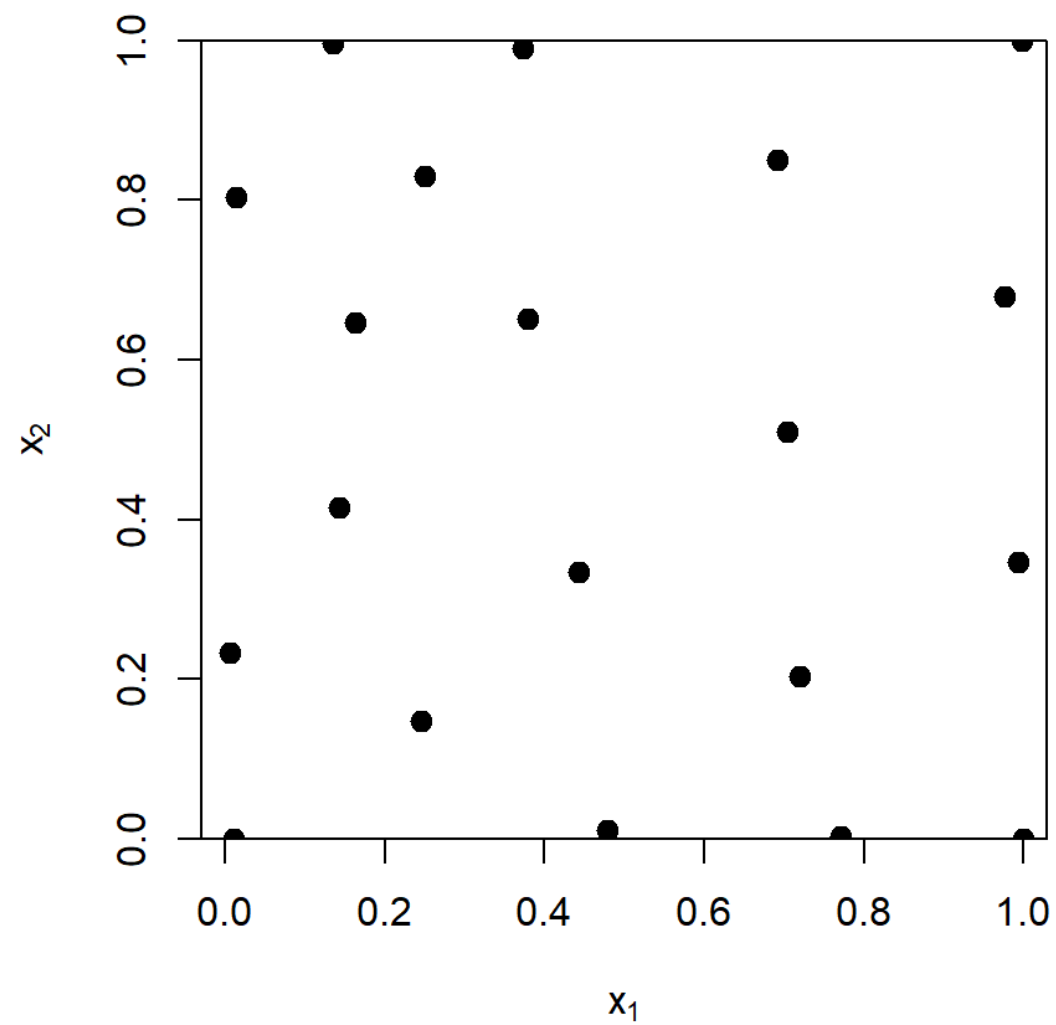
- $p = 2, n = 4$:



Simple Random



Maximin



Latin Hypercube Design

- An example: $p = 2, n = 5$:

	•		
		•	
•			
			•

$$D = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 3 \\ 4 & 1 \end{bmatrix}$$

- Each row and column has one and only one point.
- Each factor has n levels.

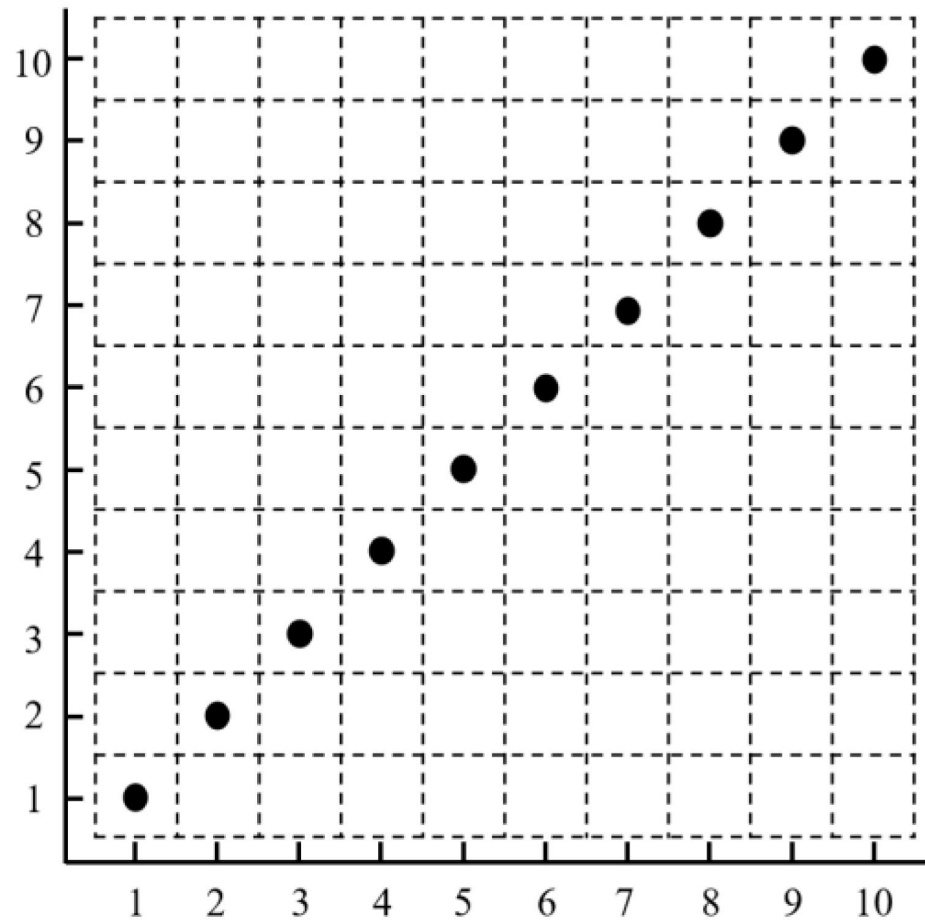
- How to construct an LHD(n, p)?

Step1 Randomly permute $\{1, 2, \dots, n\}$ for each $x_1, \dots, x_p$.

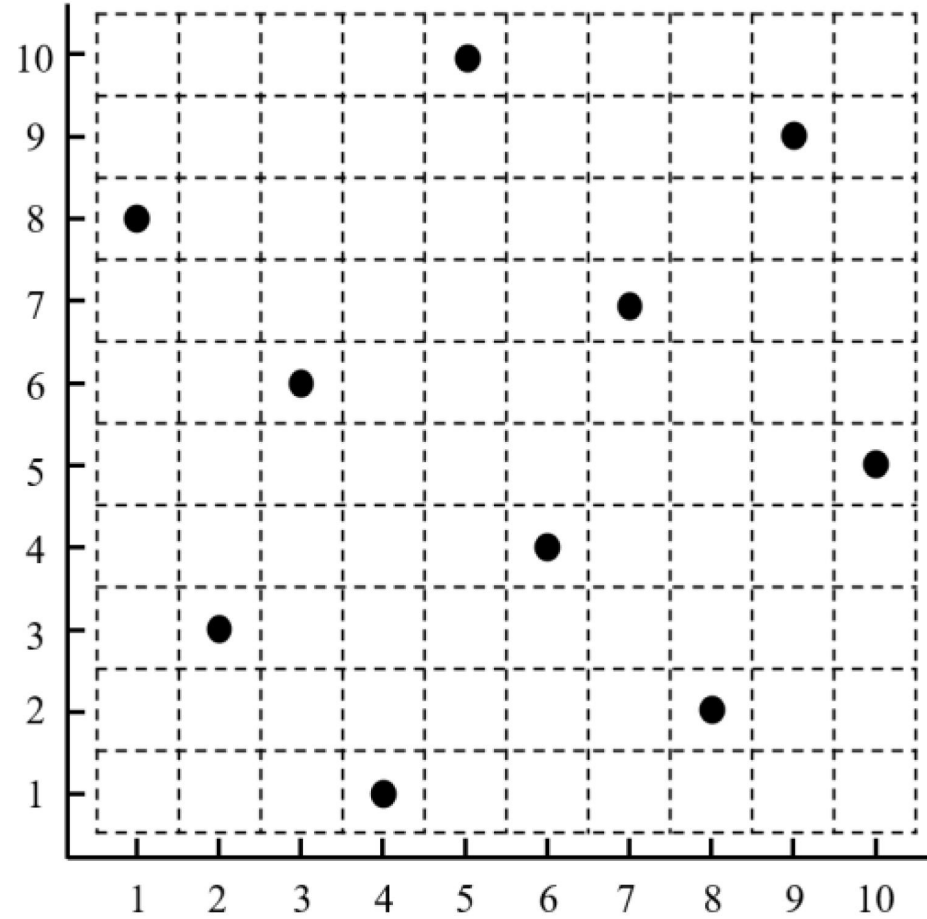
Step2 $D' \leftarrow \frac{D-0.5}{n}$, where $D \in \{1, 2, \dots, n\}^p$. Thus, $D' \in [0, 1]^p$.

$$D = \begin{bmatrix} x_1 & x_2 & x_3 \\ 1 & 2 & 7 \\ 2 & n & 9 \\ 3 & 5 & 1 \\ \vdots & \vdots & \vdots \\ n & 3 & 2 \end{bmatrix}$$

Limitation of LHD



(a) A non-space-filling LHD



(b) A space-filling LHD

Maximin / minimax LHD

Optimization problem

$$\max_{\text{LHD}} \min_{i \neq j} \|x_i - x_j\|$$

- search over LHD permutations
- keep LHD structure

Optimization problem

$$\min_{\text{LHD}} \sup_x \min_i \|x - x_i\|$$

- harder than maximin
- requires approximating the supremum